

Line 4 Sorter

Machine Documentation Package

Customer	Midwest Container Co. (<i>fictional sample customer</i>)
Plant / location	Plant 2, Davenport, IA
Machine / line	Line 4 Sorter — two-lane carton sortation with reject
Controller	Allen-Bradley CompactLogix 5370, 1769-L33ER, firmware 30.011
Source materials	LINE4_SORTER.ACD (uploaded from processor 2026-05-28 by D. Voss); Cedar Valley Systems electrical set E-101...E-114 (2011, redlined shop set); HMI screenshots (4 screens); 10 machine photos; completed intake questionnaire
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How this document was built and what it does NOT do: Everything in this package is traced from the source materials listed above. Nothing is guessed. Where the materials were ambiguous or silent, the item appears in **Appendix A — Open Questions for Plant Staff** instead of being invented. The program documented is the 2026-05-28 upload, confirmed by the plant as current.

Contents

1. Sequence of Operations
2. Logic Narrative (by routine)
3. I/O List
4. Alarm List
5. Troubleshooting Guide — Top 10 Failure Modes
6. Network & Architecture Diagram

Appendix A — Open Questions for Plant Staff

1. Sequence of Operations

Written for anyone who works around the machine — no controls background needed.

1.1 What this machine does

Finished cartons come off the Line 4 packer onto this sorter. Each carton is scanned, and based on the first letter of its barcode it is sent to **Lane A** (to the stretch wrapper) or **Lane B** (to the floor-load dock). Cartons whose barcode can't be read — or whose code isn't recognized — are pushed off into the **reject chute** partway down the line.

The machine is five conveyors in a row (infeed, metering belt, sorter belt, and the two lane takeaways), a barcode scanner, a swing-arm diverter that steers cartons to Lane A or Lane B, and an air-powered pusher that knocks rejects into the chute.

1.2 Startup

1. Close both guard doors (one at the divert section, one at the reject section), pull out all three emergency stops (operator console, discharge end, and the pull-cord along the infeed), and make sure plant air is on.
2. Turn the selector switch to **AUTO**. The green START button lights up when the machine is ready.
3. Press **START**. The horn sounds for 3 seconds — this is the legally required warning that the conveyors are about to move. Nothing moves while the horn is sounding.
4. After the horn, all five conveyors start, the green stack light comes on, and the machine is running. The scanner is active automatically.

If the START button doesn't light: a guard is open, an e-stop is in, the safety relay hasn't reset, or air pressure is low. The blue RESET button flashes when the machine needs a reset before it will start.

1.3 Normal operation — following one carton through

1. The carton arrives from the packer onto the **infeed conveyor** (it breaks the first photoeye on its way in).
2. It crosses onto the **metering belt**, which automatically stop-starts to put at least 18 inches of gap between cartons — the sorter needs that gap to track each carton individually.
3. As the carton leaves the metering belt it passes the **scanner photoeye**, which fires the barcode scanner.
4. The machine decides where the carton goes: barcode starting with "**A**" -> Lane A; "**B**" -> Lane B; anything else, or no successful read -> **reject**. The decision is remembered in a tracking queue while the carton rides the sorter belt.
5. **If it's a reject:** as the carton reaches the reject station, the pusher extends and sweeps it into the chute, then retracts before the next carton arrives.
6. **If it's a keeper:** before the carton reaches the diverter, the swing arm moves to the correct side — its resting position is Lane A; air swings it over for Lane B. A sensor on each side confirms the arm actually got there.
7. The carton rides off onto its lane takeaway conveyor and is counted on its way past the lane-entry photoeye.

1.4 When a lane backs up

If cartons back up to the end-of-lane photoeye for more than 5 seconds, that lane is "full": its beacon lights (Lane A's at the wrapper, Lane B's at the dock), the amber stack light comes on, and the alarm shows on the screen. **The line keeps running** — cartons for the other lane still sort normally. Only when **both** lanes are full does the infeed stop feeding new cartons. Lane-full alarms clear themselves as soon as the lane is worked down; no reset needed.

1.5 The "DV" keyswitch — what it actually does

The hand-labeled keyswitch on the side of the panel (asked about in the intake questionnaire) is wired to the PLC and the code is unambiguous: **when turned ON, every carton is sent to Lane B regardless of barcode**. Rejects stop happening; reads are ignored. The MANUAL screen shows this as "B-LANE MODE: ON." The rung comment, dated 2016, reads *"DV SW – RUN ALL TO B LANE – DALE."* That behavior is verified from the code. Its intended use isn't stated anywhere, but it matches the questionnaire's shrink-wrap quirk: turning this key sorts unscannable display cartons to Lane B without taping over labels. **Recommendation:** label it "ALL CARTONS TO LANE B" and add it to operator training. (Confirm the intended use with Dale; OQ-3 covers the related product-code question.)

1.6 Stopping

- **Normal stop:** press STOP. All conveyors stop, the seal-in drops out, the diverter and pusher stay where they are. Restart is the normal start sequence (horn and all).
- **Emergency stop:** any of the three e-stop devices, or opening either guard door, kills the safety relay. Air is dumped from the diverter and pusher (the dump valve is held open *by* the healthy safety circuit), all motion stops immediately, and the red light comes on. After clearing the cause: close guards / pull out the e-stop, press the flashing blue RESET to reset the safety circuit, then START as normal.

1.7 Recovering from a fault

The general pattern: the red stack light comes on, the horn pulses until acknowledged on the ALARMS screen, the screen names the alarm. Fix the cause, then press **RESET** once (momentary). Two special cases every operator should know:

- **After a sort-area jam,** clear the cartons by hand, then **hold RESET for 3 full seconds** (with the sort area photoeyes clear). This purges the tracking queue so the machine forgets the cartons you removed. If you only tap RESET, the queue is out of step with reality and cartons will mis-sort or fault again — *this is the answer to "sometimes it won't restart until Dale does something."*
 - **A sorter-belt drive fault cannot be cleared from the screen** — see failure mode 3 in the troubleshooting guide. The note taped inside the panel door ("cycle sorter disconnect") is correct, and Section 5 explains why.
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2. Logic Narrative

The program is one continuous task, one program, eight routines. **R00_Main** calls the others every scan, in this order: **R01_Safety** -> **R02_Mode** -> **R03_Infeed** -> **R04_ScanTrack** -> **R05_SortDivert** -> **R06_Lanes** -> **R07_Alarms**. A rung comment warns not to reorder — scanning must run before diverting because the tracking queue is loaded in R04 and unloaded in R05.

2.1 R00_Main — call manager

Purpose: First-scan initialization and routine calls.

What it does: On controller first scan it zeroes the carton tracking queue (**Track_FIFO**), its control structure, and all alarm bits — so a power cycle always starts with an empty, consistent tracking state. Rungs 1–7 are unconditional subroutine calls in the order above.

Signals referenced: none (internal only).

2.2 R01_Safety — safety circuit, air, reset

Purpose: Gather the safety devices into one healthy/unhealthy decision and control the air supply.

What it does: All three e-stop channels (**EStop_PB1_OK**, **EStop_PB2_OK**, **EStop_Pullcord_OK**) plus the MSR127 safety-relay feedback (**SafRly_OK**) must be healthy, then both guard doors (**Guard1_Closed**, **Guard2_Closed**), to make **Safety_OK**. Air is healthy (**Air_OK**) only when the mechanical pressure switch (**AirPressSw_OK**) agrees with the analog transducer (**AirPress_PSI** >= 65.0 psi setpoint) — a deliberate two-source check. The main air dump valve (**Sol_AirDump**) is held open by **Safety_OK**, so any safety trip dumps air from the diverter and pusher. A safety loss while running latches **SafetyTrip**, which must be cleared with **PB_Reset** after the circuit is healthy again; the blue reset lamp (**ResetPB_Lamp**) flashes at 1 Hz whenever a reset is required.

Signals referenced: **EStop_PB1_OK**, **EStop_PB2_OK**, **EStop_Pullcord_OK**, **SafRly_OK**, **Guard1_Closed**, **Guard2_Closed**, **AirPressSw_OK**, **AirPress_PSI**, **Sol_AirDump**, **PB_Reset**, **ResetPB_Lamp**.

2.3 R02_Mode — auto/manual, start/stop, stack light

Purpose: Run the start/stop station and line-level state.

What it does: The selector (**SS_Auto**) chooses AUTO; off is manual/jog. A start request needs AUTO, **Safety_OK**, **Air_OK**, no latched **SafetyTrip**, the normally-closed STOP healthy (**PB_Stop**), and START pressed (**PB_Start**). The request starts a 3-second timer (**StartWarn_T**) that sounds the **Horn** — the pre-motion warning — and only on its completion latches **Auto_Run**, the line-running seal-in. STOP, a safety loss, or leaving AUTO drops **Auto_Run** instantly. The stack light follows state: **SL_Green** running, **SL_Amber** added when a lane is full, **SL_Red** on any alarm. The START button lamp (**StartPB_Lamp**) lights steady when the machine is ready to start.

Signals referenced: **SS_Auto**, **PB_Start**, **PB_Stop**, **Horn**, **SL_Green**, **SL_Amber**, **SL_Red**, **StartPB_Lamp**.

2.4 R03_Infeed — infeed & metering conveyors, gapping

Purpose: Feed cartons in, with enforced gaps.

What it does: In auto, the infeed drive (MTR101) runs unless both lanes are full, there's a sort-area jam (**Alm[8]**), or the scanner is offline/over the no-read limit (**Alm[12]**) — the philosophy is *stop feeding* when downstream can't handle product. The metering belt (MTR102) stop-starts off the gap photoeye **PE102_Meter** with a 750 ms timer (**Meter_T**) to hold the 18-inch minimum gap noted in the rung comment. A carton sitting on the entry eye **PE101_InfeedEntry** more than 6 s while the infeed runs raises the infeed-jam half of **Alm[11]**. In manual, **PB_JogInfeed** jogs both belts at jog speed. Drive faults from either MTR101 or MTR102 set the shared **Alm[4]** "infeed section drive fault."

Signals referenced: PE101_InfeedEntry, PE102_Meter, PB_JogInfeed, MTR101 (start/speed/fault over Ethernet), MTR102 (same).

2.5 R04_ScanTrack — scanning, sort decision, carton tracking

Purpose: Decide each carton's destination and remember it until the carton reaches the diverter.

What it does: The scanner photoeye PE103_ScanTrig fires a 200 ms hardwired trigger pulse (Scn_Trigger) to the Cognex scanner; the result string comes back over Ethernet into Scn_ReadString. Destination: first character "A" -> Lane A, "B" -> Lane B, anything else or a no-read -> reject. **The DV keyswitch (SS_DV) overrides everything: ON forces every carton to Lane B** (Dale's 2016 addition — see §1.5). On the trailing edge of the scan eye, the destination is pushed into the tracking queue (Track_FIFO, FFL instruction). The routine also keeps a rolling no-read percentage over the last 100 reads; above the 5.0% setpoint, or if the scanner heartbeat is lost for 5 s, it raises Alm[12], which stops the infeed. Finally, the jam-recovery purge lives here: **holding PB_Reset 3 s with both sort-area eyes clear empties the tracking queue** (rung comment: "HOLD RESET 3 SEC TO PURGE TRACKING AFTER JAM - DV").

Signals referenced: PE103_ScanTrig, Scn_Trigger, SS_DV, PB_Reset, PE104_SortTrack, PE105_SortConfirm (purge permissives), SCN101 (read string, good-read, heartbeat over Ethernet).

2.6 R05_SortDivert — diverter, reject pusher, sorter belt, jam detection

Purpose: Physically execute each carton's sort decision.

What it does: When a carton breaks the tracking eye PE104_SortTrack, its destination is pulled from the queue (FFU). Lane B energizes the diverter solenoid Sol_DivertB; Lane A de-energizes it (the arm is spring-returned to A). A 2 s confirm timer then expects the matching position prox (DivPrx_LaneA / DivPrx_LaneB) — no confirmation raises Alm[6]. Rejects arm the pusher instead: at the reject window (PE110_RejConfirm) the pusher solenoid Sol_RejPush extends; the extend prox (RejPush_Extended) must answer within 1.5 s and the retract prox (RejPush_Retracted) must be made again before the next carton, else Alm[7]. Either sort-area eye (PE104/PE105_SortConfirm) blocked over 4 s while the sorter runs is a jam: Alm[8], and **the line hard-stops** (Auto_Run unlatched — rung comment: "JAM = HARD STOP, CLEAR BY HAND"). The sorter belt (MTR103) runs with Auto_Run absent a jam; PB_JogSorter jogs it in manual. An MTR103 drive fault sets Alm[5] — and notably, **no fault-reset command is programmed for MTR103** (the other four drives get one in R07). The rung comment says "525 FLT RESET - 103 MISSING, ASK DALE." This is why a sorter drive fault can't be cleared from the screen and the crew cycles its disconnect (OQ-5). A display-only counter (TrackErr_Ct, DIAG screen) accumulates mismatches between confirmed cartons and queue unloads — a health indicator for the tracking system.

Signals referenced: PE104_SortTrack, PE105_SortConfirm, PE110_RejConfirm, DivPrx_LaneA, DivPrx_LaneB, RejPush_Extended, RejPush_Retracted, Sol_DivertB, Sol_RejPush, PB_JogSorter, MTR103 (start/speed/fault over Ethernet).

2.7 R06_Lanes — takeaway conveyors, lane-full, counting

Purpose: Run the two discharge lanes and watch for backups.

What it does: Both takeaways (MTR104, MTR105) run whenever the line runs. The end-of-lane eyes (PE107_LaneA_Full, PE109_LaneB_Full) blocked for 5 s set the lane-full alarms (Alm[9]/Alm[10]) and light the corresponding beacon (LaneA_FullBeacon at the wrapper, LaneB_FullBeacon at the dock). These alarms self-clear when the eye unblocks. The lane-entry eyes (PE106_LaneA_Entry, PE108_LaneB_Entry) drive throughput counters shown on the MAIN screen. Drive faults set Alm[2] (Lane A) and Alm[3] (Lane B).

Signals referenced: PE106_LaneA_Entry, PE107_LaneA_Full, PE108_LaneB_Entry, PE109_LaneB_Full, LaneA_FullBeacon, LaneB_FullBeacon, MTR104, MTR105 (start/fault over Ethernet).

2.8 R07_Alarms — alarm management, horn, resets

Purpose: Collect alarm conditions, run the horn and red light, and handle resets.

What it does: Safety-circuit problems (e-stops, safety relay, and guard doors — guards are folded into the same bit via `SafetyTrip`) set `Alm[1]`. A blocked reject chute (`PE111_RejFull`, 3 s debounce) sets `Alm[11]` — **the same bit as an infeed jam**; the HMI text reads "INFEED JAM / REJECT FULL" to cover both (OQ-7). The low-air alarm rung writes its bit *indirectly* through a pointer tag that currently equals 4 — meaning **low air turns on `Alm[4]`, which displays as "INFEED SECTION DRIVE FAULT"** (OQ-6, and failure mode 6 in the troubleshooting guide). Any newly active alarm pulses the `Horn` (1 s on/off) until acknowledged from the HMI and holds `SL_Red` solid. A momentary `PB_Reset` clears latched alarms 2–8 if their cause is gone, and issues a fault-clear command to four of the five drives — MTR103's is missing (see §2.6). Lane fulls self-clear; the scanner alarm clears on reset once the no-read rate is back under the setpoint and the scanner is online. The reset lamp (`ResetPB_Lamp`, flashed from R01) tells the operator a reset is wanted. Alarm bits are packed into words for the HMI alarm display.

Signals referenced: `PE111_RejFull`, `PB_Reset`, `Horn`, `SL_Red`, `ResetPB_Lamp`, MTR101/102/104/105 fault-clear (Ethernet).

3. I/O List

3.1 Hardwired I/O — 50 points (42 wired, 8 spare)

Address	Tag	Description	Type	Drawing ref
Local:1:I.0	PB_Start	Start pushbutton, operator console (green, illuminated)	DI	E-106
Local:1:I.1	PB_Stop	Stop pushbutton, operator console (red, N.C.)	DI	E-106
Local:1:I.2	PB_Reset	Fault reset pushbutton (blue, illuminated)	DI	E-106
Local:1:I.3	SS_Auto	Auto/Manual selector (1 = Auto)	DI	E-106
Local:1:I.4	PB_JogInfeed	Jog infeed PB, panel (manual mode)	DI	E-106
Local:1:I.5	PB_JogSorter	Jog sorter PB, panel (manual mode)	DI	E-106
Local:1:I.6	EStop_PB1_OK	E-stop channel — operator console	DI	E-102
Local:1:I.7	EStop_PB2_OK	E-stop channel — discharge end	DI	E-102
Local:1:I.8	EStop_Pullcord_OK	E-stop channel — infeed pullcord (relocated 2014 per redline)	DI	E-102
Local:1:I.9	SafRly_OK	MSR127 safety relay aux feedback	DI	E-102
Local:1:I.10	Guard1_Closed	Guard door switch — divert section	DI	E-102
Local:1:I.11	Guard2_Closed	Guard door switch — reject section	DI	E-102
Local:1:I.12	AirPressSw_OK	Main air pressure switch (field-set; setpoint unknown — OQ-1)	DI	E-104
Local:1:I.13	DivPrx_LaneA	Diverter arm prox — Lane A position	DI	E-105
Local:1:I.14	DivPrx_LaneB	Diverter arm prox — Lane B position	DI	E-105
Local:1:I.15	RejPush_Retracted	Reject pusher prox — retracted	DI	E-105
Local:2:I.0	PE101_InfeedEntry	Photoeye — infeed entry from packer	DI	E-105
Local:2:I.1	PE102_Meter	Photoeye — metering belt gap eye	DI	E-105
Local:2:I.2	PE103_ScanTrigger	Photoeye — scanner trigger	DI	E-105
Local:2:I.3	PE104_SortTrack	Photoeye — sorter belt tracking eye	DI	E-105
Local:2:I.4	PE105_SortConfirm	Photoeye — post-divert confirm	DI	E-105
Local:2:I.5	PE106_LaneAEntry	Photoeye — Lane A entry (count)	DI	E-105
Local:2:I.6	PE107_LaneA_Full	Photoeye — Lane A full, end of lane	DI	E-105
Local:2:I.7	PE108_LaneBEntry	Photoeye — Lane B entry (count)	DI	E-105
Local:2:I.8	PE109_LaneB_Full	Photoeye — Lane B full, end of lane	DI	E-105
Local:2:I.9	PE110_RejConfirm	Photoeye — reject chute entry confirm (moved 2017 per redline)	DI	E-105
Local:2:I.10	PE111_RejFull	Photoeye — reject chute full	DI	E-105
Local:2:I.11	RejPush_Extended	Reject pusher prox — extended	DI	E-105
Local:2:I.12	SS_DV	"DV" keyswitch, panel side — forces all cartons to Lane B (§1.5)	DI	E-103 redline, term. 2126
Local:2:I.13–15	—	SPARE	DI	E-103

Address	Tag	Description	Type	Drawing ref
Local:3:O.0	SoL_DivertB	Diverter arm solenoid (energize = Lane B; spring = Lane A)	DO	E-104
Local:3:O.1	SoL_RejPush	Reject pusher solenoid (energize = extend)	DO	E-104
Local:3:O.2	SoL_AirDump	Main air dump valve (energized = air on, by safety circuit)	DO	E-104
Local:3:O.3	SL_Green	Stack light — green, running	DO	E-106
Local:3:O.4	SL_Amber	Stack light — amber, lane full warning	DO	E-106
Local:3:O.5	SL_Red	Stack light — red, fault	DO	E-106
Local:3:O.6	Horn	Start warning / alarm horn	DO	E-106
Local:3:O.7	Scn_Trigger	Scanner hardwire trigger	DO	E-108
Local:3:O.8	LaneA_FullBeacon	Lane A full beacon, at stretch wrapper	DO	E-106
Local:3:O.9	LaneB_FullBeacon	Lane B full beacon, at dock	DO	E-106
Local:3:O.10	StartPB_Lamp	Start PB illumination (ready)	DO	E-106
Local:3:O.11	ResetPB_Lamp	Reset PB illumination (flash = reset required)	DO	E-106
Local:3:O.12–15	—	SPARE	DO	E-103
Local:4:I.Ch0	AirPress_PSI	Main air transducer, 0–150 psi, 4–20 mA (alarm SP 65 psi)	AI	E-104
Local:4:I.Ch1	—	SPARE	AI	E-103

3.2 Network devices and key signals (EtherNet/IP)

Node	Address	Signal	Description
MTR101 Infeed (PF525, 2 HP)	192.168.4.31	O.Start/Stop, O.FreqCommand	Run command & speed (InfeedSpd_SP)
		I.Faulted	-> Alm[4]
		O.ClearFaults	Reset on PB_Reset
MTR102 Metering (PF525, 1 HP)	192.168.4.32	O.Start/Stop, O.FreqCommand	Gap stop/start from PE102_Meter
		I.Faulted	-> Alm[4] (shared with MTR101)
		O.ClearFaults	Reset on PB_Reset
MTR103 Sorter (PF525, 3 HP)	192.168.4.33	O.Start/Stop, O.FreqCommand	Sorter belt run & speed
		I.Faulted	-> Alm[5]
		(O.ClearFaults NOT programmed)	See §2.6 / OQ-5 / failure mode 3
MTR104 Lane A (PF525, 2 HP)	192.168.4.34	O.Start/Stop; I.Faulted; O.ClearFaults	Run with line; fault -> Alm[2]
MTR105 Lane B (PF525, 2 HP)	192.168.4.35	O.Start/Stop; I.Faulted; O.ClearFaults	Run with line; fault -> Alm[3]
SCN101 Cognex DataMan 280	192.168.4.40	Read string -> Scn_ReadString	Barcode result (trigger is hardwired DO above)
		Good-read bit; heartbeat -> Scn_Online	No-read tracking; offline -> Alm[12]
HMI PanelView Plus 7	192.168.4.20	HMI_AlmWord , commands, AlmAck	4 screens: MAIN / MANUAL / ALARMS / DIAG
PLC 1769-L33ER	192.168.4.10	—	Controller

4. Alarm List

#	Bit	HMI message	Cause (from logic)	Operator response
1	Alm[1]	E-STOP / SAFETY CIRCUIT OPEN	Any e-stop in, pullcord pulled, safety relay dropped, or a guard door opened (guards trip via the same latch)	Find and clear the device — check all 3 e-stops and both guard doors. Close/pull out, press flashing RESET, then START.
2	Alm[2]	LANE A TAKEAWAY DRIVE FAULT	MTR104 (192.168.4.34) reports faulted	Read the fault code on the drive face (panel, 4th drive). Press RESET — this drive accepts a remote fault clear. If it re-faults, see failure mode 10 / call maintenance with the code.
3	Alm[3]	LANE B TAKEAWAY DRIVE FAULT	MTR105 (192.168.4.35) reports faulted	Same as Lane A, 5th drive.
4	Alm[4]	INFEED SECTION DRIVE FAULT	MTR101 or MTR102 faulted — or low air pressure (a programming quirk writes the low-air alarm onto this same bit — OQ-6)	Check the air gauge first (failure mode 6). If air is fine, read codes on drives 1–2, clear cause, press RESET.
5	Alm[5]	SORTER DRIVE FAULT	MTR103 (192.168.4.33) reports faulted	RESET will not clear this one — no remote clear is programmed for this drive. Follow failure mode 3 (cycle the sorter drive's disconnect).
6	Alm[6]	DIVERTER POSITION FAULT	Arm commanded to a lane but position prox didn't confirm within 2 s	Check for a carton blocking the arm; check air pressure. Clear obstruction, RESET, restart. Recurring on cold mornings -> failure mode 2.
7	Alm[7]	REJECT PUSHER FAULT	Pusher didn't confirm extend within 1.5 s, or didn't confirm retract before the next carton	Check for a wedged carton at the chute; check air. Verify the pusher moves freely in manual. RESET.
8	Alm[8]	SORT AREA JAM	Tracking or confirm photoeye blocked > 4 s while sorter running — hard-stops the line	Remove jammed cartons by hand. Hold RESET 3 s with the sort area clear to purge tracking (§1.7), then START. Full procedure: failure mode 1.
9	Alm[9]	LANE A FULL	End-of-lane eye blocked > 5 s	Work down the lane at the wrapper. Self-clears; no reset. Both lanes full = infeed pauses.
10	Alm[10]	LANE B FULL	End-of-lane eye blocked > 5 s	Work down the lane at the dock. Self-clears.
11	Alm[11]	INFEED JAM / REJECT FULL	Two causes, one bit: carton stuck on infeed entry eye > 6 s, or reject chute full eye blocked > 3 s (OQ-7)	Check both places: the infeed entry from the packer, and the reject chute. Clear/empty, RESET.
12	Alm[12]	SCANNER OFFLINE / HIGH NO-READ	No-read rate > 5% over last 100 cartons, or scanner heartbeat lost 5 s — stops the infeed	If running display/shrink-wrapped cartons, use the DV key (all to Lane B, §1.5). Otherwise clean the scanner window, check label quality, verify scanner power/link. RESET once rate recovers.

5. Troubleshooting Guide — Top 10 Failure Modes

Ranked with the plant's stated biggest problem first. Written for a plant electrician or trained operator.

1. Carton jam at the divert point (A1m[8] SORT AREA JAM)

Symptom: Line hard-stops, red light, screen shows SORT AREA JAM. A carton is skewed/wedged at the diverter arm. The plant reports this a few times a week, worst with half-slotted cartons.

Likely causes, in order: (1) Carton clipped by the arm mid-transfer — gap too small or carton late vs. its tracking slot. (2) Under-weight/damaged carton drifting on the sorter belt. (3) Diverter arm slow (see #2) so it's still moving when the carton arrives.

Diagnostic steps: 1. Open the divert-section guard (this also drops safety — expected). 2. Remove every carton between the scanner eye and the lane entries — *all of them, not just the wedged one*. 3. Close the guard, press flashing RESET to reset safety. 4. **Hold RESET 3 seconds** — the reset lamp area / DIAG screen confirms tracking purge (queue must be emptied because you removed the cartons it was tracking). 5. START.

Fix / prevention: If jams cluster on a specific carton size, check the metering gap (750 ms `Meter_T` was tuned for 18 in — OQ-4) and the sorter belt speed setpoint. A rising `TrackErr_Ct` on the DIAG screen between jams means tracking is drifting — call your integrator/TribalKnowledge before it becomes daily.

2. Diverter slow or faulting on cold mornings (A1m[6])

Symptom: DIVERTER POSITION FAULT repeatedly during the first hour on cold days (plant reports Monday mornings worst — building coldest after the weekend setback). Alarm history screenshot confirms the Monday clustering.

Likely causes, in order: (1) Moisture/condensation in the air line freezing or slugging the cylinder — check the filter/regulator bowl for water. (2) Cylinder lubrication stiff when cold. (3) Solenoid valve sticking (the "Dale vs. everyone" debate — the valve model is OQ-2). (4) Prox switch gap drifted, marginal make.

Diagnostic steps: 1. On a cold morning, watch one cycle: does the arm *move but arrive late* (air/lube — causes 1–2) or *not move at all until the valve is tapped* (valve — cause 3)? This single observation settles the debate. 2. Drain the filter bowl; check the air dryer if there is one. 3. With air locked out, stroke the arm by hand — binding? 4. Check both prox LEDs make solidly at each end of travel.

Fix: Drain legs/auto-drain on the drop, valve replacement if cause 3, or as a stopgap raise `DivConf_T` from 2.0 s — but treat that as masking, not fixing.

3. Sorter drive fault won't clear from the screen (A1m[5])

Symptom: After a power blip or fault, SORTER DRIVE FAULT stays on. RESET clears everything else but not this. Crew ritual: cycle the sorter drive's disconnect. Dale's note inside the panel door says the same.

Root cause (from the program — this is the answer to questionnaire item 8): The reset rung sends a fault-clear command to four of the five drives. **MTR103's clear command was never programmed** — the rung comment literally says "525 FLT RESET - 103 MISSING, ASK DALE." Cycling the disconnect power-cycles the drive, which clears the fault the hard way. The ritual works; it was never magic.

Fix: One rung addition (copy the MTR104 pattern to MTR103) ends the ritual permanently. Until then: open the disconnect for the 3rd drive (192.168.4.33), wait 30 s, re-close, RESET, START. Flagged as OQ-5 in case Dale had a reason.

4. High no-read rate / everything rejecting (A1m[12])

Symptom: Reject chute filling fast, infeed pauses, SCANNER alarm. Worst on shrink-wrapped display cartons.

Likely causes, in order: (1) Shrink-wrap glare defeating the scanner — a known limitation here. (2) Dirty scanner window. (3) Label print quality/placement drift from the packer. (4) Scanner offline (this alarm covers both — check DIAG for "scanner online" status).

Diagnostic steps: 1. DIAG screen: is the scanner online? If offline -> check power and the Ethernet link at the switch, port for .40. 2. If online: look at the last 10 rejects — same product? 3. Wipe the window. 4. Check `NoRead_Pct` live on DIAG while running known-good cartons.

Fix: For display/shrink-wrap runs, **turn the DV key** — that is what it's for (§1.5) — and turn it back OFF afterward (cartons sort to B regardless of label while it's on). For chronic no-reads on a normal product, the label spec or scanner tuning needs attention.

5. Lane full stops production (`Alm[9]` / `Alm[10]`)

Symptom: Amber light, beacon at the wrapper (A) or dock (B). If both lanes fill, infeed stops and the packer backs up.

Likely causes: (1) Downstream genuinely behind (wrapper down, dock crew pulled away). (2) End-of-lane photoeye blocked by a leaning carton or misaligned. (3) Takeaway drive faulted (`Alm[2]`/`Alm[3]` will also be up).

Diagnostic steps: 1. Look down the lane — actually full? Then it's a downstream staffing/equipment issue, not this machine. 2. If the lane is *not* full but the alarm is on: check the end-of-lane eye for blockage/alignment (reflector knocked askew is common). 3. Check the lane drive status on DIAG.

Fix: Self-clears once the eye unblocks. No reset needed — tell operators not to hit RESET reflexively here; it does nothing for this alarm.

6. "INFEED SECTION DRIVE FAULT" but the drives are fine (`Alm[4]` — actually low air)

Symptom: `Alm[4]` on screen; drives 1 and 2 show no fault on their displays. Confusing — until you know.

Root cause (from the program): The low-air alarm rung writes through a pointer that lands on bit 4 — **low air pressure displays as INFEED SECTION DRIVE FAULT** (OQ-6).

Diagnostic steps: 1. Look at the air gauge / DIAG pressure readout first — below 65 psi? That's your alarm. 2. Check the plant air supply, the lockout valve, the filter/regulator. 3. Only if air is good, then read the drive displays.

Fix: Restore air; alarm clears with RESET. Permanent fix: repoint the low-air alarm to an unused bit with its own HMI message — a 15-minute change we recommend.

7. Won't start: START button not lit, no alarm obvious

Symptom: Operator presses START, nothing, horn never sounds.

Likely causes, in order: (1) Selector not in AUTO. (2) Flashing blue RESET ignored — a safety trip is latched and must be reset first. (3) A guard door looks closed but its switch isn't made. (4) Air below the pressure switch's cut-in (`AirPressSw_OK` — and note the analog must *also* read ≥ 65 psi). (5) STOP button stuck/wiring (it's N.C. — a broken wire looks like STOP pressed).

Diagnostic steps: 1. Selector to AUTO. 2. If RESET is flashing, press it. 3. DIAG screen shows every permissive: e-stops, guards, relay, air — find the one that's off. 4. Push/pull each guard door firmly. 5. Check the gauge.

Fix: Per cause. If DIAG shows all permissives healthy and it still won't arm, check `PB_Stop` continuity — N.C. circuit.

8. Reject pusher slow, weak, or faulting (`Alm[7]`)

Symptom: Rejects nudged but not cleared into the chute, or REJECT PUSHER FAULT.

Likely causes: (1) Low/marginal air (see #6 — and remember low air may already be showing as `Alm[4]`). (2) Flow controls drifted or cylinder seals worn — slow stroke misses the 1.5 s confirm window. (3) Extend/retract prox out of position. (4) Carton weight changed (heavier product than the timing was tuned for — note the pusher timing was

modified by Dale per the questionnaire).

Diagnostic steps: 1. Cycle the pusher in manual mode and watch: full stroke? Crisp? 2. Check both prox LEDs at each end. 3. Check air at the pusher's drop, not just the main gauge. 4. Compare stroke time to the 1.5 s window.

Fix: Adjust flow controls/repair cylinder; realign proxies. If product weight changed permanently, the confirm preset may genuinely need retuning — document the change.

9. E-stop reset doesn't take (A_{lm}[1] stays on)

Symptom: Everything is pulled out/closed, RESET pressed, alarm stays.

Likely causes, in order: (1) One of the *three* e-stop devices missed — the pullcord especially (it latches at the switch and must be reset *at the cord*, both sides tensioned). (2) A guard door switch not made (door warped, switch loose). (3) MSR127 safety relay itself — check its LEDs (monitored-circuit fault won't reset until both channels agree). (4) Wiring — the 2014 pullcord relocation splice (per redline E-102) is a known historical disturbance point.

Diagnostic steps: 1. Walk all three e-stops: console, discharge end, pullcord — reset the pullcord at its switch. 2. Check both guard switches engage. 3. Open the panel: MSR127 LED states (both channels + reset). 4. DIAG screen shows each channel individually.

Fix: Per device. If the MSR127 shows a channel fault repeatedly, megger/inspect the pullcord run first given its splice history.

10. Line dead or erratic with drives "missing" (network loss)

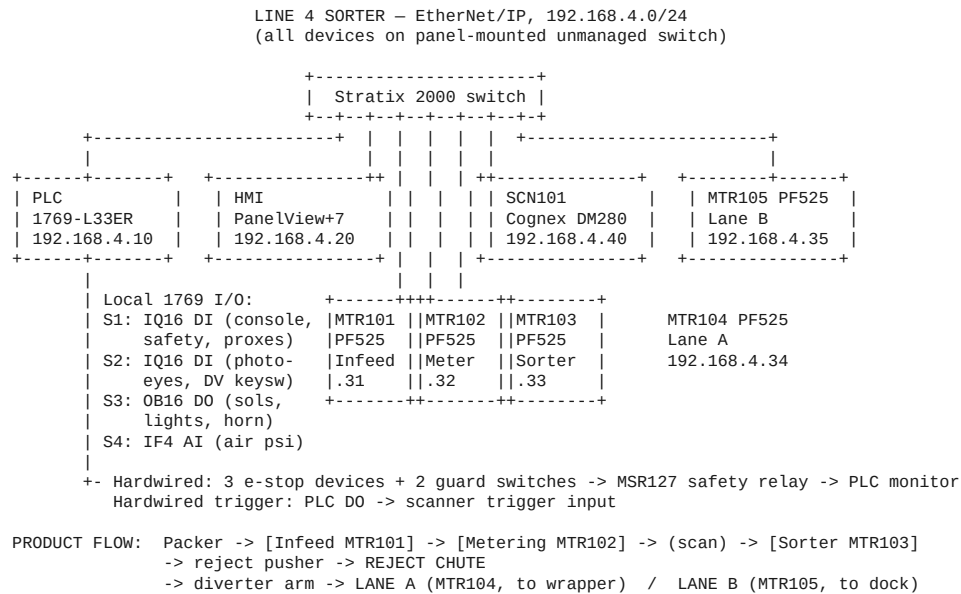
Symptom: Drives won't start with no drive fault shown, HMI values frozen or stale, possibly multiple drive-fault alarms at once after a recovery.

Likely causes: (1) Ethernet switch power or a failed port (unmanaged Stratix 2000 — no diagnostics, so it hides). (2) A damaged patch cable (panel door pinch point). (3) IP conflict from a temporarily connected laptop/device.

Diagnostic steps: 1. Link LEDs at the switch: PLC .10, HMI .20, drives .31–.35, scanner .40 — all should blink. 2. Reseat/replace suspect cables. 3. If one device only: ping it from a laptop on the panel switch (addresses in Section 6). 4. After any switch power cycle, expect drive comm faults — RESET clears them (except MTR103: failure mode 3).

Fix: Replace cable/switch as found. Recommendation: the unmanaged switch is a single point of failure with zero diagnostics; a managed replacement would tell you which port dropped.

6. Network & Architecture Diagram



What breaks if each link is lost: PLC<->drives — conveyors won't start or stop at the drive's comm-loss setting; PLC<->scanner — everything rejects, then **Alm[12]** stops the infeed; PLC<->HMI — machine *keeps running* but blind: no alarms visible, no acknowledge. The safety circuit is fully hardwired and is not affected by any network failure.

Appendix A — Open Questions for Plant Staff

Things the source materials could not answer. Fifteen minutes with Dale before March closes most of this list — answers go into revision B at no charge.

#	What we observed	The question	Who probably knows
OQ-1	Air pressure switch (E-104) is field-set; the dial is unreadable in photo 9. The analog alarm setpoint is 65 psi.	What is the mechanical switch's cut-in/cut-out set to? It should coordinate with the 65 psi analog setpoint.	Dale
OQ-2	Diverter and pusher solenoid valve part numbers are illegible on the E-104 scan; not visible in photos.	Valve makes/models, for spares stocking — and it settles the cold-morning "solenoid vs. air" debate if one was ever replaced.	Dale / storeroom records
OQ-3	Sort logic: only first-character "A" and "B" route to lanes; <i>everything</i> else rejects. The DV key bypasses to Lane B.	Are there current or planned product codes besides A/B? If a third family exists, today it all rejects unless someone turns the key.	Production scheduling
OQ-4	Metering gap timer is 750 ms (rung comment says 18 in minimum gap). Jams reported worst on half-slotted cartons.	Has carton mix changed since this was tuned? The gap timing may need re-tuning for the current sizes.	Dale / production
OQ-5	No fault-reset command programmed for the sorter drive MTR103; comment says "ASK DALE."	Was this deliberate (e.g., forcing an inspection after a sorter fault), or just never finished? We recommend adding it unless there was a reason.	Dale
OQ-6	The low-air alarm writes through a pointer tag landing on bit 4, colliding with INFEED SECTION DRIVE FAULT.	Confirm no one depends on this behavior, then re-point low air to its own bit and HMI message.	Dale
OQ-7	Alm[11] is shared: infeed jam and reject-chute-full raise the same alarm with a combined HMI text.	Do operators know to check both locations? Splitting into two bits/messages is a small change worth making.	Operators / Dale
OQ-8	No-read alarm setpoint is 5.0% with no description and no record of why.	Is 5% the right threshold for the current product mix, or a 2011 default nobody revisited?	Dale / quality

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